

Type 1, But Not Type 2, Diabetes Reduces Resistance to Damage Accumulation in Femoral Cortical Bone in Older Adults

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INTRODUCTION: Diabetes mellitus (DM) is associated with an increased risk of fracture compared to non-diabetics, with a 5-fold increased risk for hip fracture in type 1 diabetes (T1D) and a 30% increased risk of hip fracture in type 2 diabetes (T2D) [1]. However, the factors contributing to skeletal fragility in DM are not well understood. Thus, the aim of this study was to assess the ability of femoral cortical bone to resist damage accumulation extracted from the cadaveric femora of older adults with T1D, T2D, and non-diabetic controls.

METHODS: We acquired whole femora post-mortem from the Joslin Medalist Study [2,3], a unique cohort of individuals with long-duration T1D (≥ 50 years) ($n = 23$). For the T2D ($n = 21$) and non-diabetic control ($n = 19$) groups, we obtained femora of similar age-, race- and sex from a commercial tissue bank. Those with end stage renal disease, lower limb amputation, or history of bone metastases were excluded. We performed impact microindentation (Osteoprobe, Active Life Scientific) on the anterior diaphysis to quantify the average indentation distance into the bone normalized to the indentation distance measured on a reference material (polymethylmethacrylate), called the Bone Material Strength index (BMSi). Then, we extracted cortical beams (2x2x40mm) from the midshaft. We used micro-computed tomography (microCT, μ CT40, Scanco Medical, 10 μm^3 isotropic voxel) to measure tissue mineral density (Ct.TMD, mg HA/cm³) and cortical porosity (Ct.Po, %) in the cortical beams. Next, we conducted damage accumulation testing, where the beams were cyclically loaded in 4-point bending with incrementally greater displacements (i.e., damage cycles; from 75% to 200% yield displacement in 25% increments). Following each damage cycle, we calculated tangent stiffness and percent residual stiffness relative to baseline. As some of the beams failed before the test ended, we have also reported the number and percentage of failed beams at each damage level per group. We employed Kruskal-Wallis or Fisher's tests to compare bone outcomes between T1D, T2D, and controls, as appropriate, with Dunn post-hoc and Bonferroni correction to adjust for multiple comparisons (significance $\alpha \leq 0.05$). Additionally, we used Spearman's rank correlations to investigate the relationship between Ct.TMD and damage accumulation. Data are presented as median (IQR).

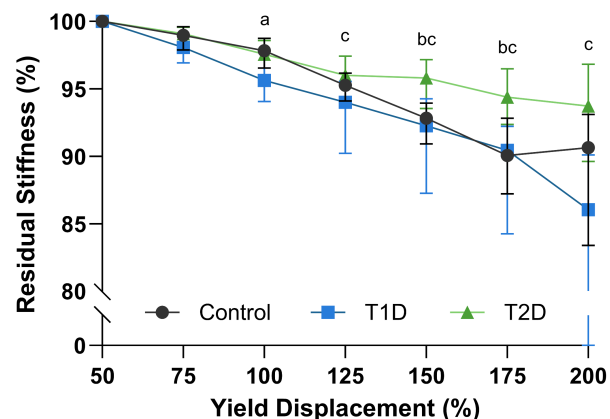
RESULTS: We obtained 24 T1D (58% women), 22 T2D (52% women), and 21 non-diabetic control (52% women) specimens. Among the T1D group, median HbA1c = 7.5 (7.1, 8.3) %, disease duration = 67 (63, 71) years (range 56-77), and age at onset = 12 (7, 15) years. While age at death (mean = 80 years) and height (mean = 1.68 m) did not differ between groups ($p > 0.8$), weight trended towards being greater in T2D versus T1D ($p = 0.069$) and control ($p = 0.087$). BMSi did not differ among groups ($p = 0.416$). However, Ct.TMD was lower in T2D versus control ($p = 0.001$) and T1D ($p = 0.078$). Cortical bone in T1D accumulated more damage, while T2D accumulated less damage, than controls (Figure). Residual stiffness was lower in T1D versus control at 100% yield displacement ($p < 0.033$) and versus T2D at 125-200% yield displacement ($p < 0.04$). Interestingly, residual stiffness was greater in T2D compared to control at 150% ($p = 0.025$), 175% ($p = 0.005$) and 200% ($p = 0.155$) yield displacement. Although no specimens failed in the control group, 29% of T1D specimens failed at or before 200% yield displacement. At 200% yield displacement, the failure rate was greater in T1D versus control ($p = 0.031$) and versus T2D ($p = 0.102$). Ct.TMD was negatively correlated with the residual stiffness at 200% yield displacement ($r_s = -0.28$, $p = 0.022$).

DISCUSSION: These results suggest that T1D altered the ability of cortical bone to resist damage accumulation. Conversely, T2D appeared to improve cortical bone resistance to damage. This differential result in T1D and T2D may be explained in part by the lower Ct.TMD in T2D and the negative correlation between Ct.TMD and residual stiffness. However, T1D and T2D may influence the physicochemical properties of the bone matrix at a different length scale than evaluated here. Thus, while the impaired resistance to damage accumulation in T1D may help to explain increased fracture risk in T1D, the unexpected improvement in damage accumulation in T2D emphasizes the need for evaluation at multiple length scales to provide a clearer understanding of the underlying factors contributing to fracture risk in this population.

SIGNIFICANCE/CLINICAL RELEVANCE: Although the growing issue of skeletal fragility in patients with DM is widely recognized, there is a lack of information regarding how DM increases fracture risk. The current observations suggest that the mechanical implications of T1D and T2D manifest differentially at the tissue level in cortical bone.

REFERENCES: [1] Vilaca et al., Bone, 2020. [2] Keenan et al., Diabetes Care, 2007. [3] Maddaloni et al., Acta Diabetologica, 2017. [4] Starup-Linde et al., Diabetes Metab Syndr Obes, 2019. [5] Nyman et al., J Orthop Res., 2007. [6] Wang et al., Bone, 2002. [7] Hunt et al., JBMR, 2019.

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Failure rate (%)	Control	T1D	T2D
Control	0	0	0
T1D	0	0	4
T2D	0	0	0
p-value	-	-	>0.999

Figure: [Top] Residual stiffness due to damage accumulation. Data presented as median $\pm$ 95% confidence intervals with Kruskal-Wallis and Dunn post-hoc test results (depicted by lower-case letters: a = $p < 0.05$ T1D versus control; b = $p < 0.05$ T2D versus control; c = $p < 0.05$ T1D versus T2D). [Bottom] Cortical beam failure rate. P-values from Fisher's test; bold = $p < 0.05$.